pfi

Python for Fixed Income

COURSE 2 · BOND MATH

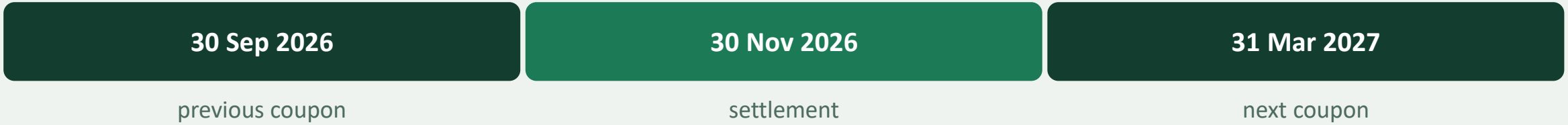
Day Counts and Accrued Interest

WEEK 2 · DAY 2

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Who earns the coupon



The seller held it: 2 months

Earned that interest, but will not be paid it

The buyer holds it: 4 months

Collects the whole coupon on 31 March

At settlement the buyer pays the seller the accrued interest, on top of the price.

```
accrued = coupon_pct / frequency x (days accrued) / (days in the period)
```

How to count those days is set by the bond's documents, not by the calendar.

Two counts, two answers

0.879808

calendar days: 61 of 182, per 100

0.875000

30/360: 60 of 180, per 100

How many days is a convention, not a fact of the calendar.

Each bond's documents name its day count. The course's bonds, like US corporate bonds generally, are 30/360.

Quorvane, 5.25 percent, 30 Nov 2026

One coupon	**5.25 / 2 = 2.625**
Excel's accrued	**0.875000**
On 1,000,000 face	**8,750.00 dollars**
Calendar, more by	**48.08 dollars**

The four US 30/360 rules

```
Days = 360 x (Y2 - Y1) + 30 x (M2 - M1) + (D2 - D1)
```

Rule	Change
1	Start is the last day of February: D1 = 30
2	Start and end both last day of February: D2 = 30
3	D1 is 31: D1 = 30
4	D2 is 31 and the start's own day is 30 or 31: D2 = 30

Case	Dates	No rule	Rules	Excel
No rule	2026-08-16 to 2026-09-30	44	44	COUPDAYBS
Rule 1	2027-02-28 to 2027-03-01	3	1	COUPDAYBS
Rule 2	2027-02-28 to 2028-02-29	361	360	YEARFRAC
Rule 3	2026-08-31 to 2026-09-30	29	30	COUPDAYBS
Rule 4	2026-07-30 to 2026-10-31	91	90	COUPDAYBS
Not rule 4	2026-10-15 to 2026-10-31	16	16	COUPDAYBS
Own day	2027-02-28 to 2027-03-31	33	31	COUPDAYBS

Every rule is about the end of a month. Rule 4 asks about the start date's own day.

days_30_360, and a lookalike

```
# rule 1: the last day of February counts as the 30th
if start_is_feb_end: d1 = 30
# rule 2: both dates on the last day of February
if start_is_feb_end and end_is_feb_end: d2 = 30
# rule 3, then rule 4 on start.day, not d1
if start.day == 31: d1 = 30
if end.day == 31 and start.day >= 30: d2 = 30
```

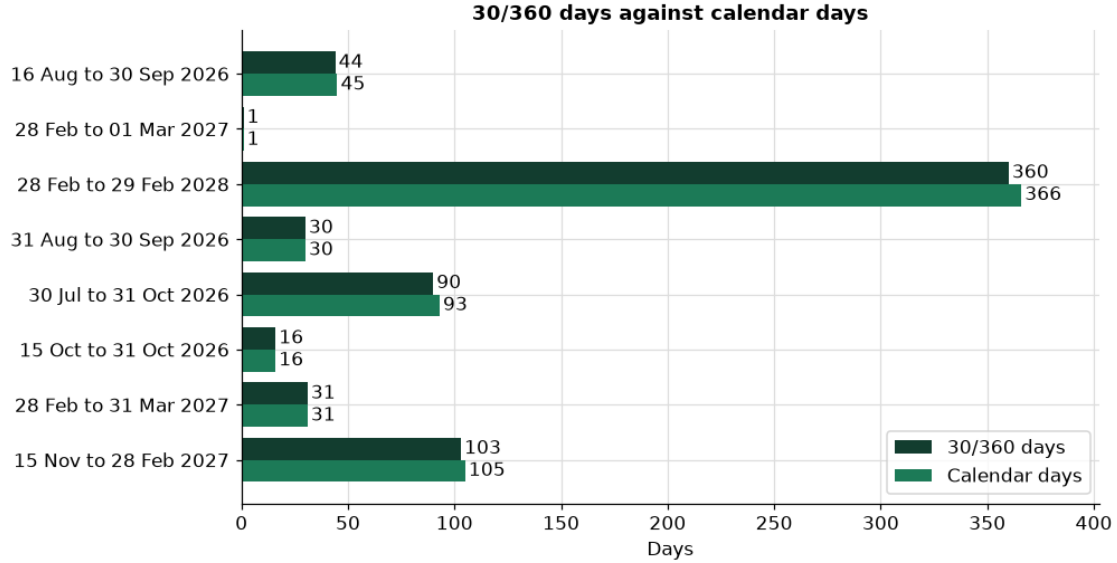
The notebook's function writes each rule as an if block with its comment; the logic is the same.

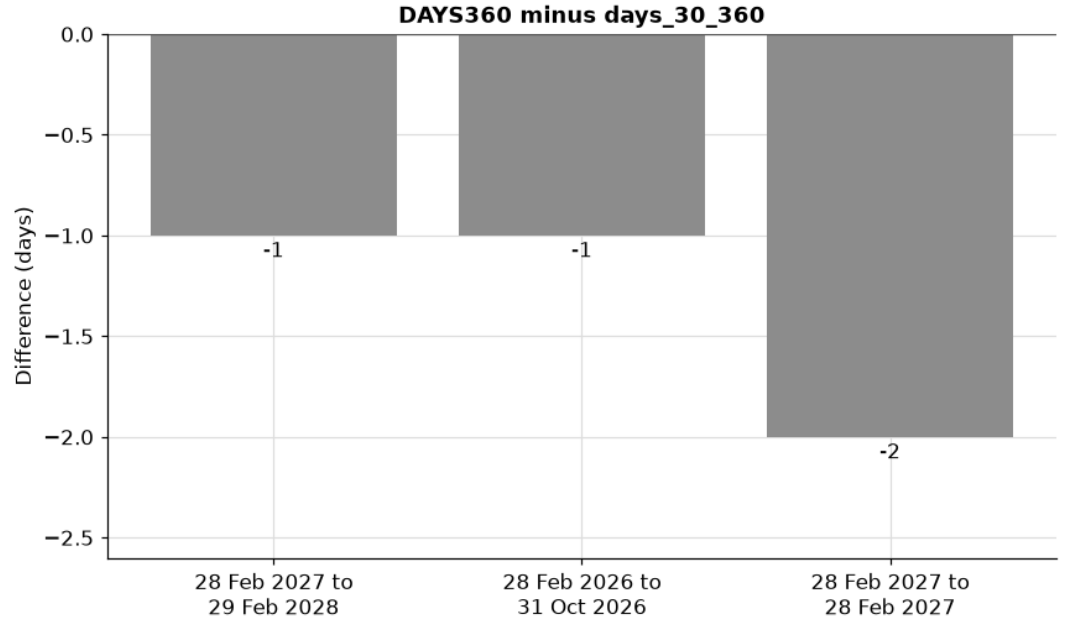
Excel side by side (tested in Excel)

`=YEARFRAC(DATE(2027,2,28),DATE(2028,2,29),0)*360`	360
`=DAYS360(DATE(2027,2,28),DATE(2028,2,29))`	359
`=YEARFRAC(DATE(2026,2,28),DATE(2026,10,31),0)*360`	241
`=DAYS360(DATE(2026,2,28),DATE(2026,10,31))`	240
`=DAYS360(DATE(2027,2,28),DATE(2027,2,28))`	-2

DAYS360 has its own February rules. The course uses the count COUPDAYBS and PRICE use.

The 30/360 counts, drawn





Close in mid-month, apart near month ends: up to 6 days in a February-to-February year.

DAYS360 is 1 or 2 days short of the count COUPDAYBS uses on all three pairs.

Actual/actual: every count in calendar days

61 / 182

actual/actual: accrued 0.879808 per 100

60 / 180

30/360: accrued 0.875000 per 100

Days accrued are calendar days, and E is the calendar length of the coupon period: 182 or 183 days for a half year.

E under actual/actual is COUPNCD - COUPPCD, the length PRICE uses.

Excel side by side (tested in Excel)

`=COUPDAYBS(DATE(2026,9,30),DATE(2031,4,30),2,1)`	153
`=COUPDAYS(DATE(2026,9,30),DATE(2031,9,30),2,1)`	181

The second is the period from 30 September 2026 to 31 March 2027: 182 calendar days. COUPDAYS with basis 1 differs from the period's length in 12 of the 28 basis 1 rows.

A, E, and DSC: coupon_period_days

Basis	A	E	DSC
30/360	60	180	120
ACT/ACT	61	182	121

A: previous coupon to settlement. E: days in the period. DSC: settlement to next coupon. Under 30/360, DSC = E - A, what PRICE uses.

Excel side by side (tested in Excel)

```
=COUPDAYBS(DATE(2026,11,30),DATE(2031,9,30),2,0)      60
=COUPDAYS(DATE(2026,11,30),DATE(2031,9,30),2,0)       180
=COUPDAYSNC(DATE(2026,11,30),DATE(2031,9,30),2,0)     120
```

Bond 99031FAF8, 30 September 2026

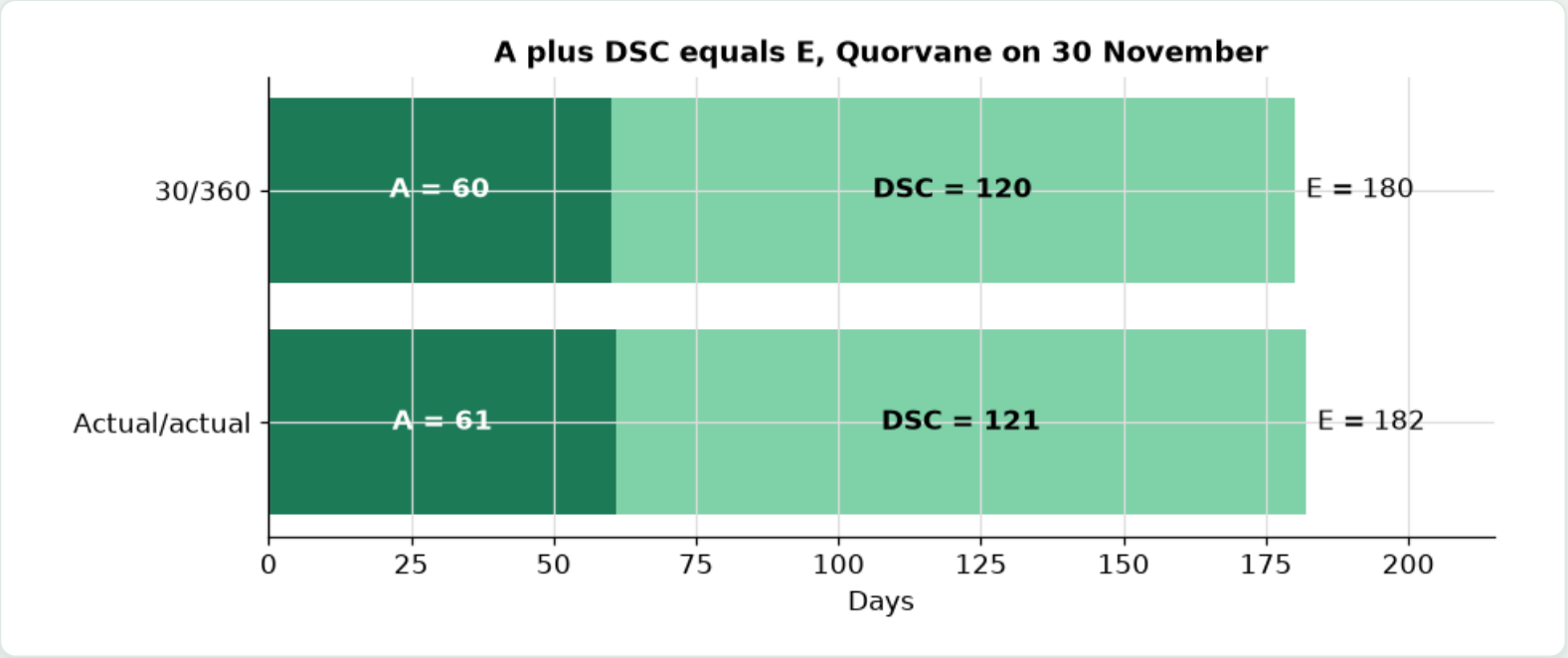
Previous coupon 2026-08-28

Next coupon 2027-02-28, a February month end

E - A	**148**	what PRICE uses
COUPDAYSNC	**150**	not used under 30/360
Calendar	**151**	actual days

Three answers to days to the next coupon. The course follows PRICE.

A, E, and DSC, drawn



Dark is A, the days accrued. Light is DSC, the days to the next coupon. Together they fill E.

Under 30/360, DSC = E - A, the count PRICE uses. Actual/actual counts calendar days throughout.

accrued_interest, and ACCRINT

Issuer	Coupon	Accrued per 100
Quorvane Energy	5.25	0.875000
Dunmarrow Rail	6.125	1.020833
Pellwick Paper	4.75	0.791667
Halvering Foods	7.5	1.250000

```
return coupon_pct / frequency * days_accrued / days_in_period
```

Where ACCRINT differs: 99031FAF8

Next coupon 28 February 2027 is a month end, so ACCRINT counts from 31 August: 33 days, 0.7448. The bond's own schedule: 32 days, 0.7222.

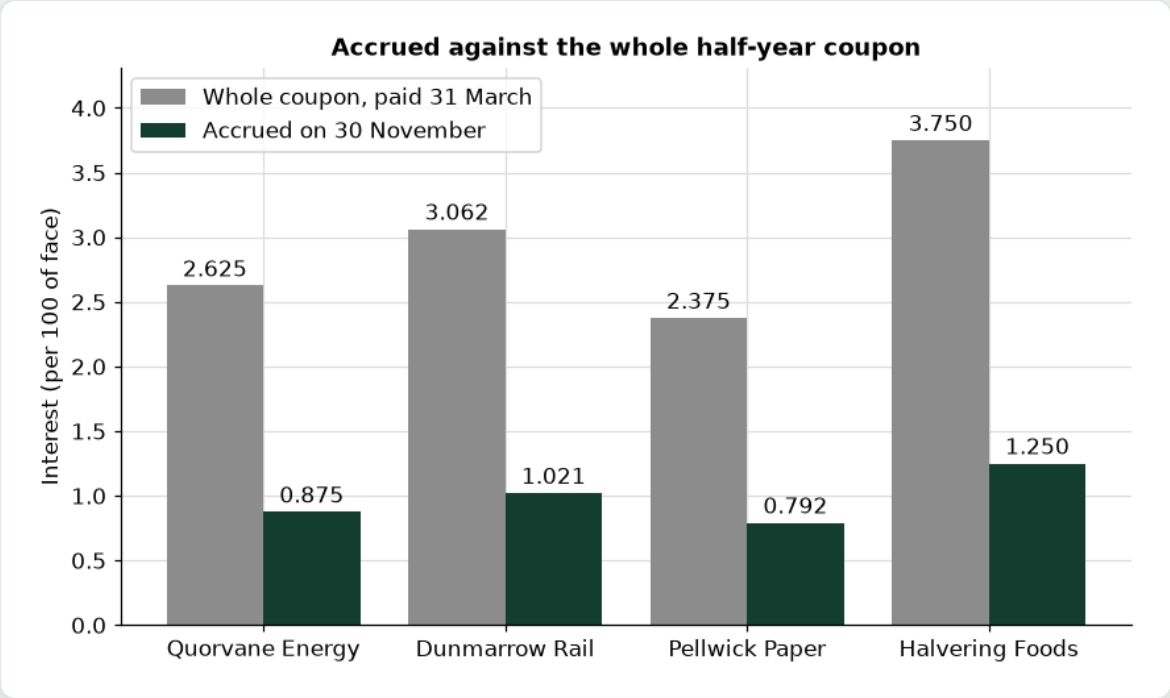
Excel side by side (tested in Excel)

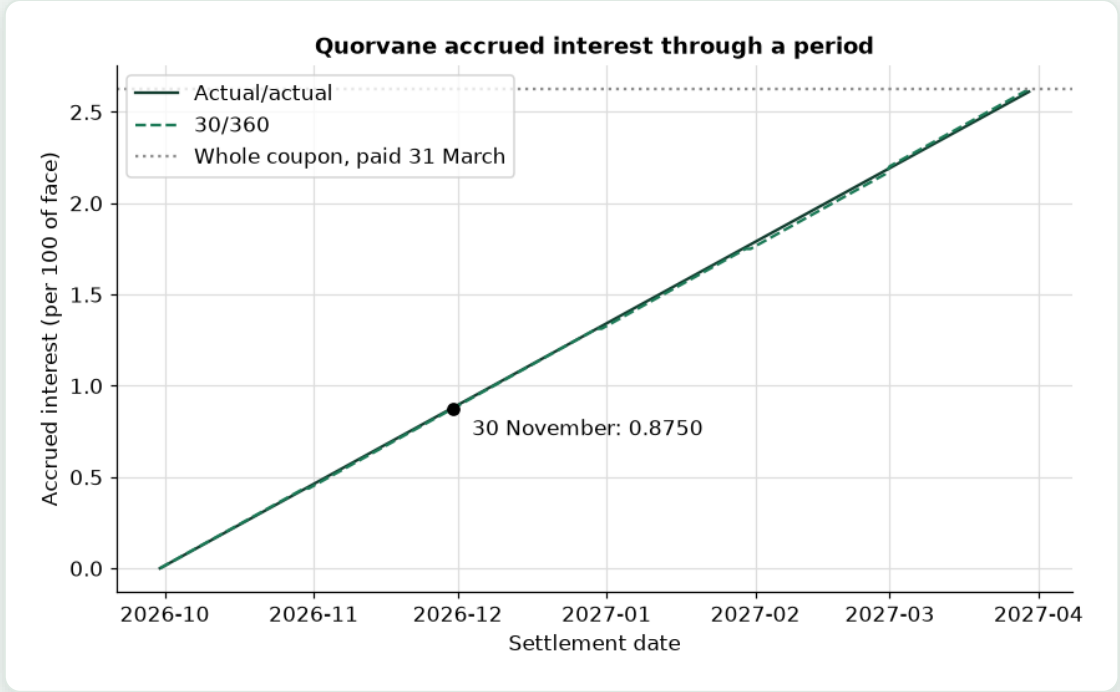
```
=100*5.25/100/2*COUPDAYBS(D)/COUPDAYS(D)                    0.875
=ACCRINT(COUPPCD(D),COUPNCD(D),DATE(2026,11,30),
   5.25/100,100,2,0)                                        0.875
```

D stands for the four COUP arguments: DATE(2026,11,30),DATE(2031,9,30),2,0

The course follows the COUPDAYBS formula, the way PRICE counts. ACCRINT is a second check.

Accrued interest, drawn





Two months of a six-month period: each bond has accrued a third of its coupon.

Accrued builds from zero to the whole coupon. The two day counts part by a hair and meet at the coupon.

The holdings accrued column, reproduced

200

bonds, every one within rounding of the file's 3 decimals

0.0005000000

largest gap per 100: half of the third decimal

7

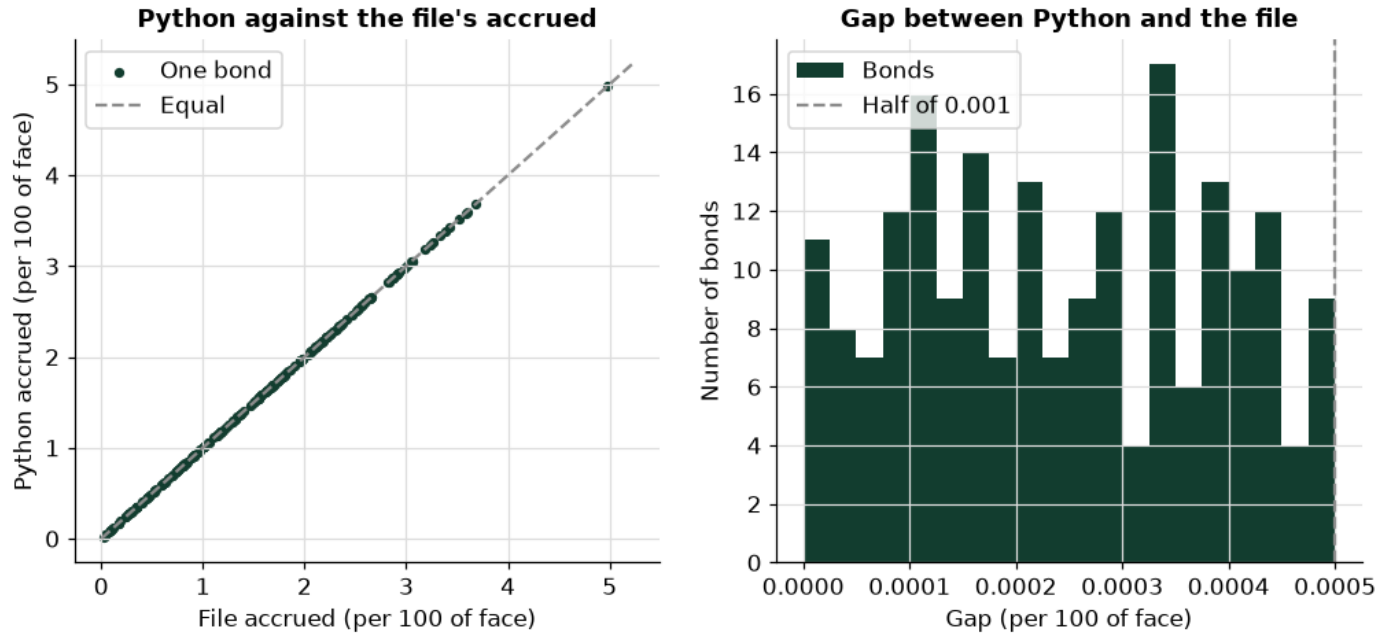
bonds at a rounding tie, such as 0.0875 filed as 0.087

So the check allows 0.0005 + 1e-9 on the unrounded value: a hair of floating point sits on top of every tie.

99031FAF8 is in the file at 0.722: the COUPDAYBS route, not ACCRINT's 0.7448.

Against Excel's unrounded values in the reference file, all 200 agree within the test's 1e-9.

The holdings column, drawn



Left: 200 dots on the line. Right: every gap is under 0.0005, the file's rounding.

Today's Excel, every formula tested

Job	Excel (tested), Quorvane on 30 Nov 2026	Returns
Days accrued, A	`=COUPDAYBS(DATE(2026,11,30),DATE(2031,9,30),2,0)`	60
Days in period, E	`=COUPDAYS(DATE(2026,11,30),DATE(2031,9,30),2,0)`	180
Days to next	`=COUPDAYSNC(DATE(2026,11,30),DATE(2031,9,30),2,0)`	120
Accrued per 100	`=100*5.25/100/2*COUPDAYBS(...)/COUPDAYS(...)`	0.875
Second route	`=ACCRINT(COUPPCD(...),COUPNCD(...),...,5.25/100,100,2,0)`	0.875
30/360 days	`=YEARFRAC(DATE(2027,2,28),DATE(2028,2,29),0)*360`	360
A lookalike	`=DAYS360(DATE(2027,2,28),DATE(2028,2,29))`	359

The /100 is in every formula: Excel takes the coupon as a decimal, bondmath in percent.

Review a suggestion by its word diff

An assistant "tidies" coupon_period_days: a year has 365 days, so use 365 // frequency. It runs.

git diff bondmath.py

```
-       days_in_period = 360 // frequency
+       # a year has 365 days
+       days_in_period = 365 // frequency
```

git diff --word-diff bondmath.py

```
{+# a year has 365 days+}
days_in_period = [-360-]{+365+} // frequency
```

The comment is true of the calendar and beside the point: the bond is 30/360. The tests fail on the first reference row.

Read the word diff, run the tests, then git restore bondmath.py: the committed file is back.

Bring it home: paste, test, read, commit

1

Your folder

Set-Location to my_bondmath, in your projects folder.

2

Paste

Today's three functions and three tests at the end of your files, without the %%writefile line.

3

Test

python -m pytest -q: 18 passed, or 24 with the exercises of days 1 to 6.

4

Read, commit

git diff --word-diff bondmath.py, then git add and git commit.

5

The habit

Any suggested edit: word diff, tests, and git restore if it is wrong.

```
git diff --word-diff bondmath.py
```

In Colab: download bondmath.py and test_bondmath.py from the work folder under /tmp. They replace yesterday's downloads.

The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.

1

Docstring

Units, convention, and the Excel formula: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert against Excel's reference rows. Run it: it fails.

→

4

Fix

Find the convention it broke, fix it, run the test again.

Today: accrued counts actual days over 365 for a 30/360 bond.

Code that runs is not the same as code that is right.

What you can do now

- Say why the buyer pays accrued interest: one coupon times A / E
- Apply the four US 30/360 rules, and say why DAYS360 differs
- Count a coupon period on actual/actual
- Return A, E, and DSC, and accrued, matched to Excel on every row
- Review a suggested change with git diff --word-diff, and restore it

Week 2, Day 3 Clean and dirty price: the quote, the invoice, and the price between coupons